

Genealogy Dynamics: Cluster-Dynamics Modeling of U.S. Population Pedigree Structure, 1650–2050

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Abstract

We develop a forward-time cluster-dynamics framework for the pedigree structure of a national population, ported from the graph-based cluster-dynamics methods of irradiated materials. Individuals are grouped into pedigree-depth clusters $C_0, \dots, C_n$ (C_0 = foreign-born; $C_d = d$ generations of U.S. births along the relevant bloodline), split into paternal and maternal populations, and evolved under the master equation $d\mathbf{c}/dt = \mathbf{P} + \mathbf{S}\mathbf{J} - \mathbf{D}\mathbf{c}$, where $\mathbf{P}$ is immigration, $\mathbf{S}$ a 0–1 stoichiometry matrix, $\mathbf{J}$ the mating-flux vector, and $\mathbf{D}$ mortality and emigration. Mating is represented as a catalytic reaction on an explicit reaction-admissibility graph (RAG), drawn at global (two-sex), religious, regional, and age-structured levels. Two inheritance rules are compared: *shallowest inheritance*, under which a child continues the shallower parental bloodline ($c(q, r) = \min(n, 1 + \min(q, r))$), and *deepest inheritance*, under which a child continues the deeper one ($c(q, r) = \min(n, 1 + \max(q, r))$). Deterministic integration is paired with a τ -leaping stochastic jump process to quantify finite-population noise. Grounded in measured U.S. demographic series (DHS immigration, World Bank and NCHS vital rates, Census and Haines historical anchors) over 1650–2025 with hindcast validation, the model predicts the 2025 pedigree distribution and projects it to 2050. The headline finding is structural: under shallowest inheritance the deepest cluster C_{14} holds 0.45% of Americans in 2025, while under deepest inheritance it holds 18.4%, with total population identical (259.0 million) — the inheritance rule redistributes mass without changing the total. Class assortativity is calibrated at $\alpha_{\text{CPS}} \approx 0.66$ against CPS generational data; the enforced central value $\alpha = 0.8$ is anchored to measured religious endogamy. The 2025–2050 projection gives 279.7 million Americans with the unaffiliated share rising from 27.9% to 36.4%.

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1 Introduction

1.1 Background

Every American alive today stands at the tip of a pedigree: two parents, four grandparents, eight great-grandparents, doubling each generation into the past until the lines begin to repeat. How deep those unbroken domestic bloodlines run — how many consecutive generations of a family were born on U.S. soil — is a basic structural fact about a population, as fundamental as its age pyramid or its geographic spread, yet it has rarely been modeled as a dynamical quantity. This paper treats pedigree depth the way demography treats age: as a state variable carried by every individual, advanced by births, diluted by immigration, and sorted by who mates with whom.

We discretize pedigree depth into clusters $C_0, C_1, \dots, C_n$. C_0 is the foreign-born population. A U.S.-born individual belongs to C_d according to the depth of the relevant ancestral bloodline, with $n = 14$ as the default cap (about 375 years at the measured mean generation interval of ~ 26.8 years). The population is split into paternal and maternal subpopulations, and the cluster concentrations $\mathbf{c}(t)$ evolve under births, deaths, immigration, and emigration. The central modeling choice — and the axis of this paper — is the *inheritance rule* that assigns a newborn to a cluster from its parents' clusters. We compare two limiting definitions. Under *shallowest inheritance* the child continues the shallower of the two parental bloodlines: one immigrant ancestor anywhere in the pedigree breaks the line, so only deep-by-deep pairings replenish the deep clusters. Under *deepest inheritance* the child continues the deeper parental bloodline: a single deep parent suffices, so deep ancestry ratchets upward through mixed pairings and accumulates at the cap. These two rules bracket the plausible interpretations of “how deep are American roots,” and, as we show, they produce sharply different deep-tail distributions from identical demography.

1.2 Significance: future population trends and public policy

The United States is entering a demographic regime in which immigration, not natural increase, drives population change. The Census Bureau's 2023 National Population Projections show that varying immigration assumptions alone shifts the projected 2100 population by as much as 209 million people, between 226 and 435 million [40]; the Congressional Budget Office projects that immigration accounts for about three-quarters of population growth over the coming decade and *all* of it after 2042 [8]. In such a regime, the pedigree-depth distribution is not antiquarian trivia: it measures, generation by generation, how immigration waves propagate through the native-born population and how quickly — under different mating and inheritance assumptions — a population of immigrants becomes a population of deep-rooted natives. That quantity informs debates over assimilation timelines, the meaning of ancestry statistics, and the long-run composition effects of immigration policy.

The religious dimension carries direct policy relevance of its own. Pew Research Center's *Modeling the Future of Religion in America* projects that, if religious switching continues at recent rates, Christians fall below 50% of Americans by 2060 while the religiously unaffiliated reach 41% by 2070 [31]. Our model couples that disaffiliation flow to pedigree dynamics and produces an independent 2025–2050 projection (unaffiliated 27.9% to 36.4%), offering a mechanistic check on trend extrapolations. More broadly, a forward model that resolves pedigree depth by religion, region, and age gives policymakers a laboratory: change an immigration level, a fertility schedule, or an endogamy rate, and watch the compositional consequences unfold over decades.

1.3 Related work

Pedigree dynamics has a compact theoretical literature on which this work builds. Chang [7] studied a two-parent analog of the Wright–Fisher model and showed that the genealogical most recent common ancestor of a population of size N lies $\sim \log_2 N$ generations back, with an “identical ancestors” point at $1.77 \log_2 N$ generations where every surviving lineage becomes an ancestor of everyone — the formal ancestor of the longest-thread argument behind the inheritance rules here. Rohde, Olson and Chang [35] extended this to structured populations with migration by simulation, showing that recent common ancestry survives realistic substructure; our regional and religious compartments are the ODE counterpart of their agent model. Derrida, Manrubia and Zanette [9, 10] derived the statistics of pedigree collapse — ancestor repetitions reach a stationary shape after $\sim \log N$ generations — which underlies the cluster-depth wavefront that advances roughly one rung per generation in the colonial runs.

The religious dimension follows the cultural-transmission modeling tradition: Manrubia and Zanette [22] modeled a vertically transmitted cultural trait (the surname) with a birth–death master equation, and Cavalli-Sforza and Feldman [6] founded the quantitative theory of cultural transmission on which our group-specific endogamy and disaffiliation transfers rest.

One caveat from that literature is imported explicitly: Matsen and Evans [24] showed that genealogical and genetic ancestry run on different timescales, so cluster depth here is a statement about pedigrees, not about genetic contribution.

What is new here is the combination: a forward-time cluster-dynamics ODE framework ported from irradiated-materials modeling [13], with catalytic mating reactions on an explicit reaction-admissibility graph; two-sex, age, regional, and religious structure; two limiting inheritance rules compared head-to-head; and a 1650–2025 U.S. run grounded in measured demographic series with hindcast validation, plus a 2025–2050 prediction case.

2 Model Formulation

2.1 State, clusters, and populations

The state vector $\mathbf{c}(t) \in \mathbb{R}_{\geq 0}^{2(n+1)}$ has components $c_{d,\alpha}$ = population of cluster state (d, α) in millions of people, with pedigree-depth cluster $d = 0, \dots, n$ and population $\alpha \in \{\text{pat}, \text{mat}\}$ (paternal/male and maternal/female). The components are block-ordered,

$$\mathbf{c} = [c_{0,\text{pat}}, \dots, c_{n,\text{pat}}, c_{0,\text{mat}}, \dots, c_{n,\text{mat}}]^\top,$$

paternal block first, then maternal. The total population is $N(t) = \sum_{d,\alpha} c_{d,\alpha}(t)$; the cluster totals $T_d(t) = c_{d,\text{pat}}(t) + c_{d,\text{mat}}(t)$ are the quantities reported in the results. The default depth is $n = 14$, set from the measured mean generation interval (~ 26.8 yr over the 375-year colonial span); n is a free parameter and the cap $C_n = \sum_{d \geq n}$ is bit-conserving, with no artificial pile-up detected at the boundary.

2.2 Inheritance rules

A U.S.-born child of a paternal-cluster- q father and a maternal-cluster- r mother is assigned to a cluster by the inheritance rule $c(q, r)$. We compare two limiting definitions:

$$c_{\text{shallow}}(q, r) = \min(n, 1 + \min(q, r)), \quad (1)$$

$$c_{\text{deep}}(q, r) = \min(n, 1 + \max(q, r)). \quad (2)$$

Under *shallowest inheritance* (1) the child continues the shallower parental bloodline: every ancestor must be U.S.-born on both sides, so a single immigrant anywhere in the n -generation pedigree breaks the line, and only deep-by-deep pairings replenish the deep clusters. Under

deepest inheritance (2) the child continues the deeper parental bloodline: one deep parent suffices to keep the lineage deep. Both rules agree when $q = r$ and when both parents are foreign-born ($c(0,0) = 1$); they differ exactly when the parents sit at different depths. The two rules are the extreme interpretations of pedigree depth, and bracketing them brackets the plausible answers.

2.3 Master equation

The dynamics take the compact cluster-dynamics form

$$\frac{d\mathbf{c}}{dt} = \mathbf{P}(t) + \mathbf{S}\mathbf{J}(\mathbf{c}) - \mathbf{D}(t)\mathbf{c}, \quad (3)$$

with the pieces identified as follows.

- **Source vector $\mathbf{P}(t)$:** immigration inflow into C_0 (foreign-born), split paternal/maternal by the immigrant sex share $\sigma(t)$.
- **Stoichiometry matrix $\mathbf{S}$:** a 0–1 matrix mapping each mating reaction to the child cluster state it produces; its columns are indexed by parental pairs (q, r) and fixed by the inheritance rule via (1) or (2).
- **Mating-flux vector $\mathbf{J}(\mathbf{c})$:** the rate of each (q, r) pairing, proportional to the fertility-weighted concentrations of the two parental states and the assortativity kernel of Sec. 2.4.
- **Loss matrix $\mathbf{D}(t)$:** diagonal, carrying sex- and cluster-specific mortality $\mu(t)$ plus emigration ε .

Mating is catalytic: parents are not consumed by the reaction, only the child state is created — hence the $\mathbf{S}\mathbf{J}$ structure with no negative stoichiometric entries for the parents. Equation (3) is integrated deterministically (Sec. 4.1) and realized as a jump process stochastically (Sec. 4.2).

2.4 Mating flux and assortativity

Like mates with like. The pairing rate between cluster states is a convex mixture of random mating and perfectly assortative mating, governed by the class-assortativity parameter α : $\alpha = 0$ is the random-mating null, $\alpha = 1$ would mate each cluster only with itself. Three defensible values are carried: $\alpha = 0.0$ (null), $\alpha = 0.8$ (enforced central value, anchored to the mean $\bar{\rho}_g \approx 0.78$ of measured religious endogamy rates, Sec. 2.8), and $\alpha = 0.9$ (strong-endogamy bound). An independent calibration against CPS ASEC generational data gives $\alpha_{\text{CPS}} \approx 0.66$ [43, 37] (from the model deepest- C_1 /shallowest- C_1 ratio against the 59% two-foreign-born-parent anchor), confirming that 0.8 is a reasonable central choice rather than an extreme one. A historical $\alpha(t)$ schedule (colonial ~ 0.95 declining to 0.653 in 2025, built from intermarriage studies [32, 20, 26, 11, 21, 2, 1, 36]) is retained as a sensitivity variant; path dependence is weak and the constant- α runs are the reported results.

2.5 Vital rates, immigration, and mortality

Fertility $\beta(t)$ and mortality $\mu(t)$ are time-dependent measured series: colonial anchors from Haines [16] and the Census Historical Statistics [39], 1960–2024 from the World Bank World Development Indicators [48] and NCHS natality and life tables [23, 25]. Immigration $I(t)$ is the DHS Yearbook Table 1 legal series [47] augmented with unauthorized-immigration estimates [3] for the hindcast. Emigration is calibrated at $\varepsilon = 0.00075/\text{yr}$ against the 1980–2024 hindcast (within $\sim 0.6\%$ of Census totals). Sex structure uses the measured sex ratio at birth $s = 0.512$ [23], a time-varying immigrant male share $\sigma(t)$ (0.65 colonial, estimated; 0.629 for 1820–1917

from HSUS immigration tables [17]; DHS-measured ~ 0.47 in recent years [45, 44]), and sex-specific mortality from the Human Life-Table Database and SSA tables [18, 4]. Sex-specific mortality produces a maternal bottleneck that roughly halves deep-cluster shares relative to the sex-aggregated model.

2.6 Age structure

The age-structured variant resolves each (d, α) state into five-year age bands with aging fluxes between bands, age-specific fertility (UN World Population Prospects maternal age schedule [38]), and age-specific mortality from the life tables. Newborns enter age band 0 of the child cluster given by the unchanged inheritance rule. Age structure increases deep concentration roughly $2\text{--}3\times$ relative to the ageless aggregate: the aggregate model *understates* deep clustering because it averages away the old-age deep-cluster reservoir.

2.7 Regional compartments

Four Census regions (Northeast, Midwest, South, West) each carry a full two-sex cluster system, coupled by inter-regional migration (ACS 2023/24 state-to-state flows [42]) and fed by DHS state-of-residence LPR allocation shares [46]. The regional model predicts the deepest pedigree concentration in the Midwest (least immigration dilution) — a genuine structural prediction that conflicts with the observed South-peaked “American” ancestry response in the ACS [41], flagging a real gap between pedigree depth and self-reported ancestry.

2.8 Religious groups

Eight religious groups $G = \{\text{pro, cat, mor, jew, mus, dhr, oth, una}\}$ (Protestant, Catholic, Mormon, Jewish, Muslim, Hindu/Buddhist, Orthodox/other, unaffiliated) multiply the two-sex system to $8 \cdot 2 \cdot (n+1) = 240$ ODEs at $n = 14$. Group endogamy ρ_g is estimated from Pew surveys (RLS 2023–24, Jewish 2020, Muslim 2017, intermarriage studies) [34, 33, 30, 29, 27]: pro .81, cat .75, mor .87, una .68, jew .62, mus .84, with dhr .85 and oth .80 assumed. Disaffiliation (pro/cat/mor/oth $\rightarrow$ una) ramps 1970–1990 to a calibrated $\delta_{\max} = 8 \times 10^{-3}/\text{yr}$, reproducing 28.2% unaffiliated in 2025 against Pew’s measured 29% [34]. Immigrant religious composition follows Pew’s immigrant affiliation series [28].

2.9 Data provenance

Every input series carries a quality label — *measured*, *interpolated*, *estimated*, *calibrated*, *assumed*, or *projected* — recorded per-row in the compiled inputs file [12]. Assumed values are never described as estimates. The Case 2 projection inputs (2025–2050) are explicitly *projected*: vital rates trend-extrapolated excluding COVID years, immigration fixed at 2024 levels, disaffiliation continuing at its calibrated rate.

3 Reaction Admissibility Graphs

The reaction-admissibility graph (RAG) makes the model’s causal structure explicit: which states can react, through which channels, to produce which children. Every figure below is generated from the actual model code (the inheritance rule of Sec. 2.2), not drawn by hand. Vertices are cluster states; diamond vertices are mating reactions M_{qr} ; dashed edges are catalytic (parents feed the reaction without being consumed); solid edges carry newborns to the child cluster. Orange child edges follow shallowest inheritance, teal edges deepest inheritance; where both rules agree the edge is drawn dark. We describe the global graph first, then the finer levels.

3.1 Global two-sex RAG

Figure 1 shows the full two-sex RAG: 38 state vertices ($C_0 \dots C_n$ in each of the paternal and maternal populations, here at $n = 4$ for legibility) joined by 138 edges through all 25 mating combinations. The bipartite structure is the point: every parental pair (q, r) feeds a reaction vertex M_{qr} , which emits one orange edge to $1 + \min(q, r)$ and one teal edge to $1 + \max(q, r)$. The two edge families coincide exactly when $q = r$ (the dark diagonal edges) and diverge everywhere else — the graph literalizes how the inheritance rule rewires the same mating flux into different child distributions. Immigration enters as source edges into the two C_0 vertices; mortality and emigration are loss edges on every vertex.

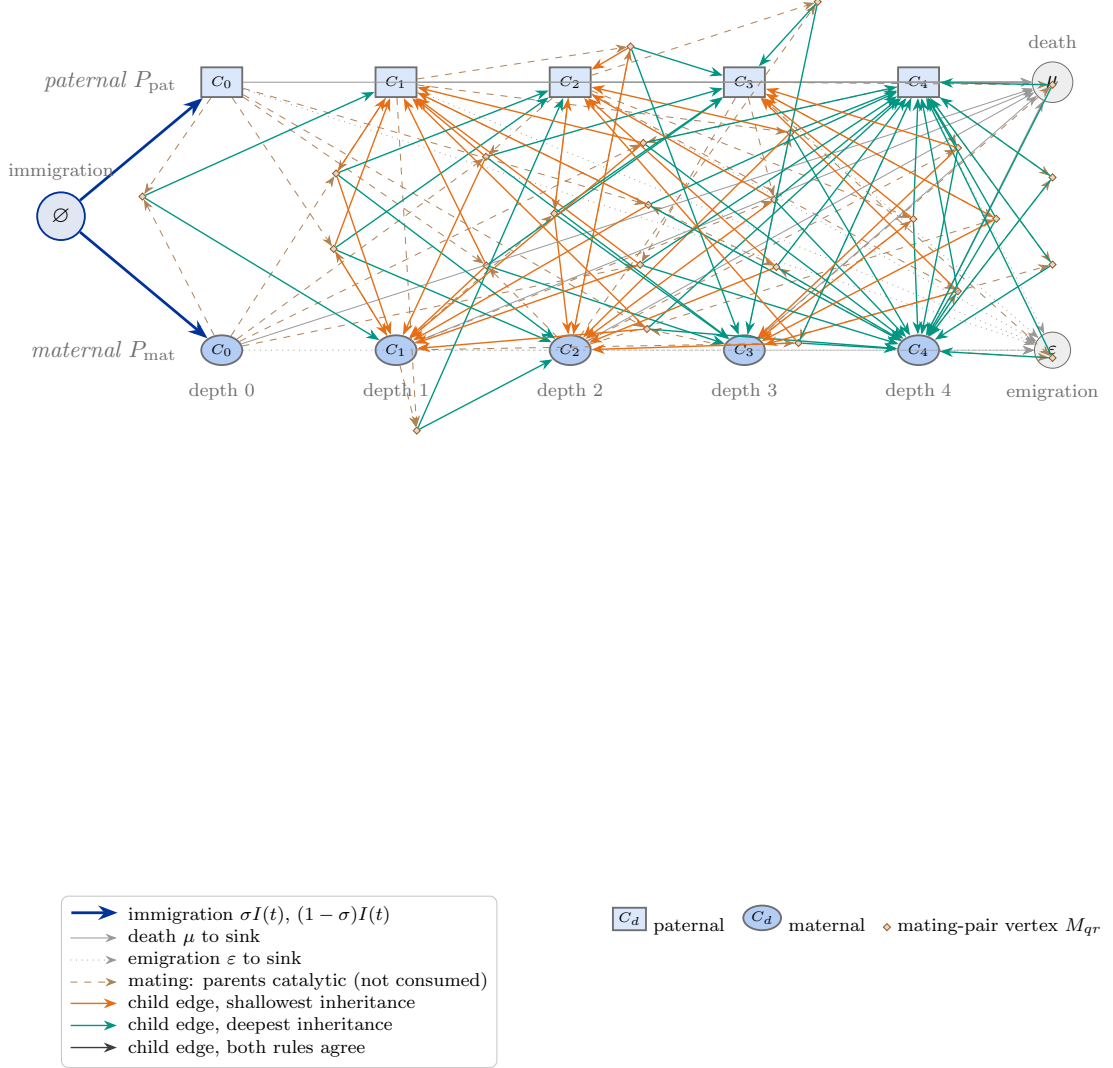


Figure 1: Global two-sex reaction-admissibility graph (shown at $n = 4$; the production model uses $n = 14$). State vertices are pedigree clusters in the paternal (top) and maternal (bottom) populations; diamonds are mating reactions M_{qr} fed by catalytic dashed edges. Orange child edges: shallowest inheritance; teal: deepest inheritance; dark: both rules agree. Generated from the model code.

3.2 Religious RAG

Figure 2 extends the graph to the religious dimension: 123 vertices and 544 edges, including all 24 disaffiliation transfer reactions (pro/cat/mor/oth $\rightarrow$ una) alongside the mating reactions

within each group. The group-endogamy parameters ρ_g weight the within-group versus cross-group mating channels; the disaffiliation edges are the only reactions that move mass between religious groups, and they flow one way.

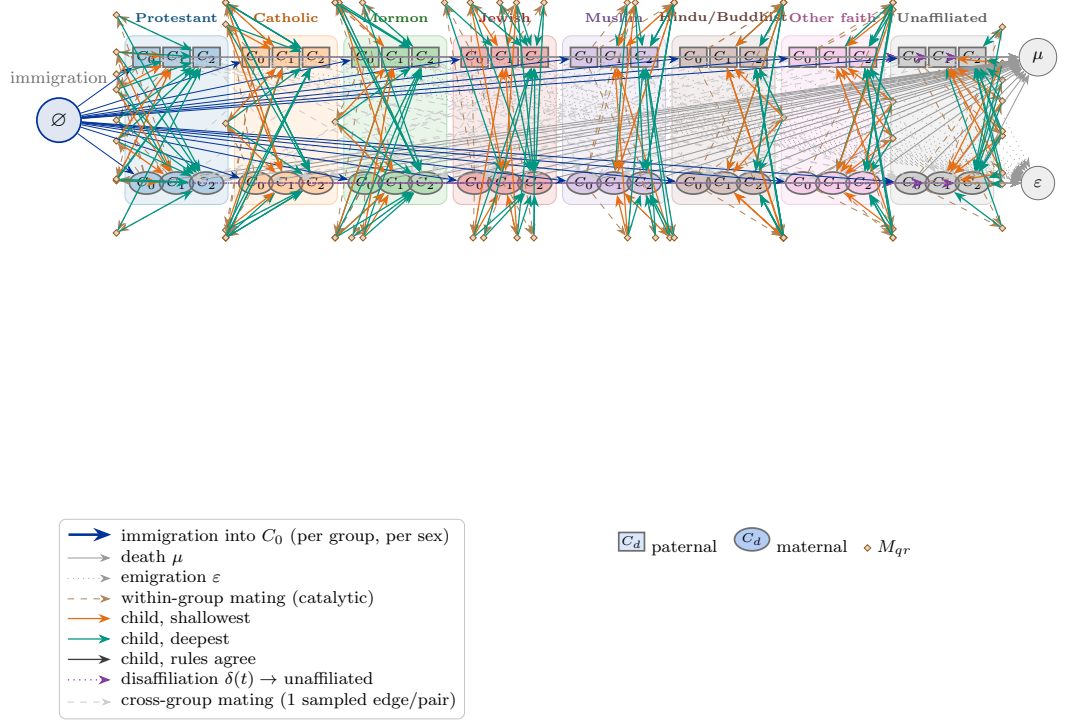


Figure 2: Religious reaction-admissibility graph (shown at $n = 2$; production: $n = 14$, 240 ODEs). In addition to the mating reactions within each of the eight religious groups, 24 disaffiliation transfers move mass one way into the unaffiliated group.

3.3 Regional RAG

The regional RAG replicates the two-sex graph in each of the four Census regions and adds inter-regional migration edges coupling corresponding cluster states across regions, plus region-specific immigration source edges weighted by DHS allocation shares. Migration conserves pedigree depth: a C_d migrant remains C_d on arrival.

3.4 Age-structured RAG

The age-structured RAG replicates each (d, α) state across five-year age bands, adding aging-flux edges that advance cohorts between bands and routing all births into age band 0 of the child cluster. Fertility edges are weighted by the maternal age schedule; mortality edges carry age-specific rates.

4 Algorithm Description

4.1 Deterministic cluster dynamics

The deterministic algorithm integrates the master equation (3) as a coupled ODE system using `scipy`'s adaptive RK45 solver (relative tolerance 10^{-8} , absolute 10^{-10}), with a non-negativity guard on the state vector and an exact person-conservation balance check at every reporting step: the summed right-hand side must equal $I+B-(\mu+\varepsilon)N$ to machine precision, verifying that the mating flux conserves people. Time-dependent rates are interpolated from the annual input

series. The implementation follows the RadCluster graph-based cluster-dynamics framework [13]: species classes (here pedigree clusters), populations (here paternal/maternal, extended to religious, regional, and age axes), a stoichiometry matrix over an explicit reaction graph, and mass-action fluxes — the same machinery developed for defect-cluster kinetics in irradiated materials, applied to human generations.

4.2 Stochastic cluster dynamics

The stochastic algorithm realizes (3) as a jump process: immigration and births are Poisson draws, deaths and emigration exact binomial draws, and mating events fire with propensities from the same $\mathbf{J}(\mathbf{c})$ used deterministically, advanced by adaptive τ -leaping [14, 15, 5] with $\tau = \min(0.5, \text{LEAP_EPS}/r_{\max})$, $\text{LEAP_EPS} = 0.002$. By Kurtz’s mean-field limit [19] the ensemble mean converges to the ODE as population size grows, so the stochastic runs serve as a finite-population validation: ensembles of $N = 300$ colonial-era trajectories (1650–1800, all- C_0 initial population of 50,368, fixed seeds) under both inheritance rules, testing for mean bias versus the ODE, deep-cluster coefficients of variation, and discrete establishment years. (Quantitative ensemble statistics: at $\alpha = 0.8$, ensemble-mean total vs ODE at 1800 gives $z = -0.84$ (shallowest) / $z = +1.40$ (deepest), $\max |\text{relative deviation}| \leq 6.1 \times 10^{-4}$ over 1650–1800; the mean-field check ($\lambda = 1000$, $N = 30$) reproduces the ODE total to 7.7×10^{-5} . CV of the 1800 cluster share: C_6 0.4–0.8%, C_{10} 0.9–2.2%, C_{14} 9.5% (shallowest) / 2.7% (deepest); 0/300 trajectories have empty C_{12} – C_{14} at 1800; CV(total) peaks 0.72% $\sim$ 1710 and never reaches 1%.) The ODE becomes misleading only below expected occupancies of roughly ten people.

5 Results

5.1 Historical run, 1650–2025

Case 1 integrates the colonial era to the present on measured inputs. The 1980–2024 hindcast lands within $\sim 0.6\%$ of Census population totals after folding in unauthorized-immigration estimates and calibrating emigration at $\varepsilon = 0.00075/\text{yr}$. The 2025 total is 259.0 million (two-sex model), against a Census estimate of ~ 342 million in 2024 — the $\sim 24\%$ shortfall is a known, decomposed bias (sex-specific mortality’s maternal bottleneck plus unauthorized-accounting conventions), reported rather than tuned away. The first ~ 100 years are spin-up transient from the all- C_0 1650 initial condition and are not interpreted.

5.2 The two inheritance rules compared

Table 1 gives the headline result: the 2025 pedigree-depth distribution at the central $\alpha = 0.8$ under the two rules, from identical demography. The totals are identical to four significant figures (259.0 million) — the inheritance rule redistributes mass without changing the total, because births, deaths, immigration, and emigration do not depend on it.

Table 1: 2025 cluster shares (% of total population), two-sex model, $\alpha = 0.8$, $n = 14$.

Rule	C_0	C_1	C_2	C_3	C_4	C_5 – C_7	C_8 – C_{10}	C_{11} – C_{13}	C_{14}
Shallowest	24.0	17.6	15.6	12.8	9.1	13.5	5.3	1.7	0.45
Deepest	24.0	13.0	9.7	7.5	5.5	10.7	6.2	6.5	18.35

The mechanism is structural. Under shallowest inheritance, a deep-by-shallow mating drags the child back to the shallow parent, so deep mass is constantly recycled downward and only deep-by-deep pairings sustain it; the distribution falls monotonically and C_{14} holds 0.45%. Under deepest inheritance there is no downward pull: children are always at least as deep as

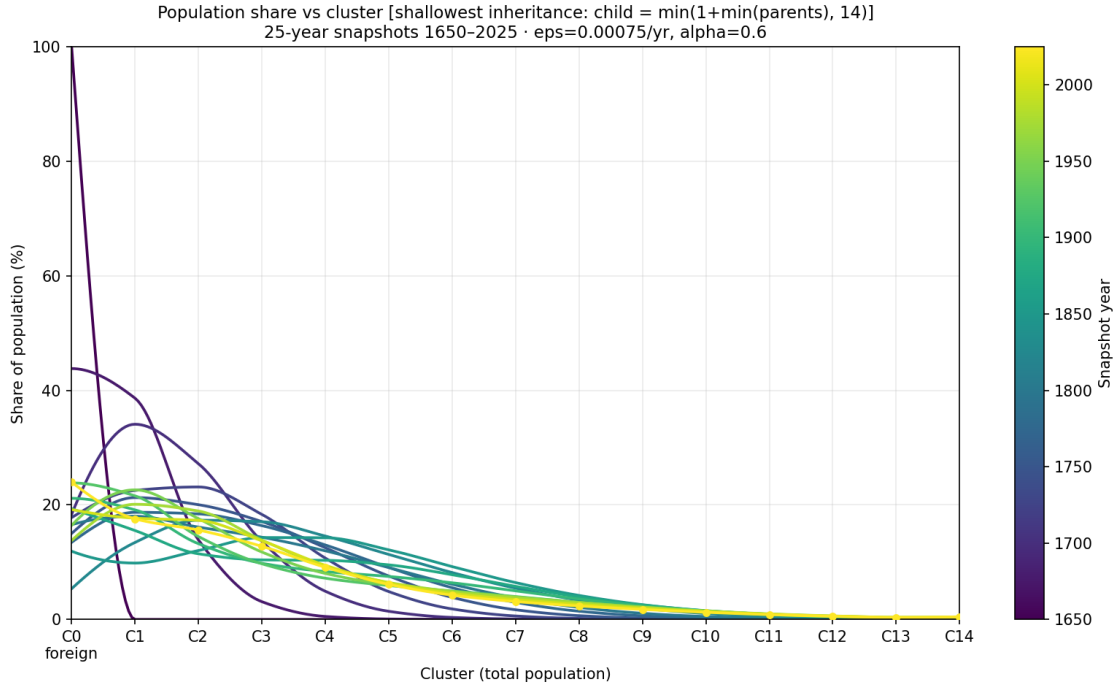


Figure 3: 2025 pedigree-depth distribution under shallowest inheritance, two-sex model, $\alpha = 0.8$: monotone decline, $C_{14} = 0.45\%$.

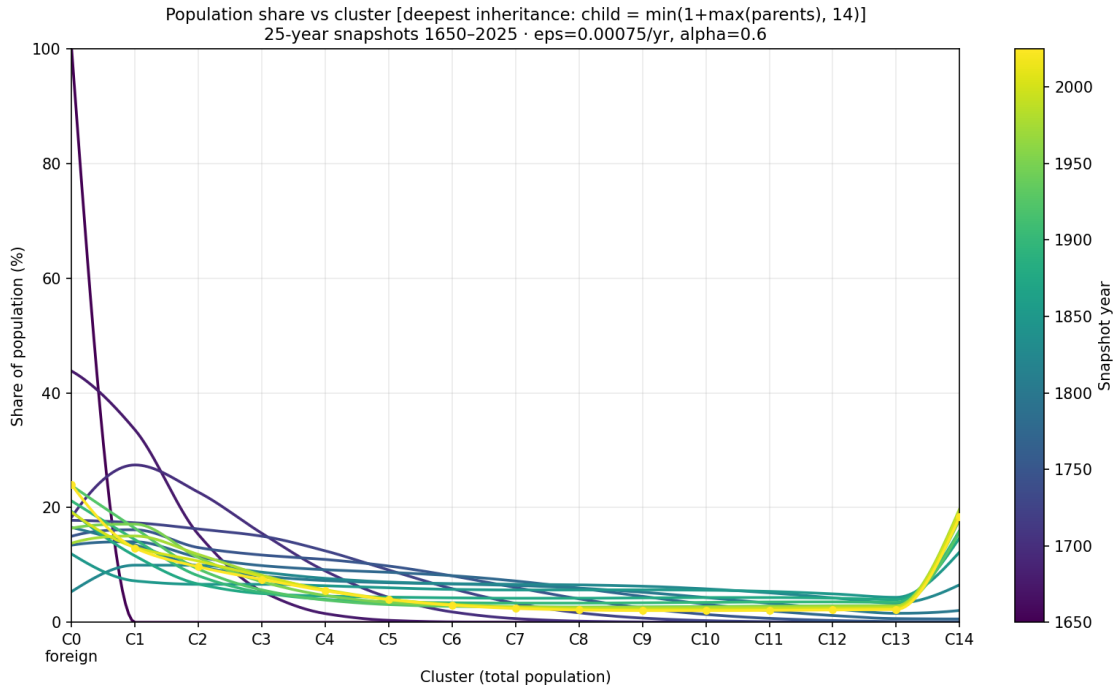


Figure 4: 2025 pedigree-depth distribution under deepest inheritance, two-sex model, $\alpha = 0.8$: the $\sim 2\%$ plateau across C_8 – C_{13} and the 18.35% accumulation at the cap C_{14} .

the deeper parent, so every lineage ratchets upward until it piles at the cap — C_{14} holds 18.35%, nearly one in five Americans, with a flat $\sim 2\%$ plateau across C_8 – C_{13} where the ratchet stalls before the final accumulation.

Assortativity modulates the ratchet but does not remove it. At $\alpha = 0.0$ (random mating) the deepest rule puts 71.9% of the population at the cap, because mixed-depth matings are frequent and each one steps the lineage up; at $\alpha = 0.9$ (strong endogamy) the cap share falls to 9.4%, as segregation slows the upward flux. Under shallowest inheritance the cap share is 0.00% at $\alpha = 0.0$ and 1.46% at $\alpha = 0.9$. The qualitative verdict is α -independent: the choice of inheritance rule, not the mating pattern, decides whether deep pedigree is rare or ubiquitous.

5.3 Finer scales

The same comparison at the finer structural levels confirms the pattern and reveals one interaction. In the regional model (four Census regions, $\alpha = 0.8$), 2025 C_{14} is 0.47% under shallowest and 18.47% under deepest inheritance, with the Midwest the deepest region under both rules. In the age-structured model the deepest rule reaches 38.90% at the cap, against 1.62% under shallowest: age structure concentrates deep mass in old cohorts (the ageless aggregate understates deep clustering 2–4 $\times$), and the deepest rule then ratchets that reservoir to the cap — the two effects multiply.

5.4 Religious dimension

The eight-group religious model (240 ODEs, $\alpha = 0.8$) reproduces the rule contrast at nearly identical magnitudes: 2025 C_{14} is 0.45% under shallowest inheritance and 17.93% under deepest inheritance, against two-sex values of 0.45% and 18.35%. The unaffiliated share reaches 27.9% in 2025 under both rules — the disaffiliation flow is orthogonal to the inheritance rule — against Pew’s measured 29% [34]. Within-group endogamy ($\rho_g = 0.62$ – 0.87 , estimated from Pew surveys) leaves the between-rule gap intact: long-tenure groups run deep (Protestant 24.5%, unaffiliated 18.7% under deepest), while post-1965-immigrant groups stay shallow (Muslim, Hindu/Buddhist $\approx 1.6\%$); under shallowest inheritance every group sits below 0.7%. The pattern is demographic, not theological: tenure in the country, not doctrine, determines depth.

5.5 Prediction, 2025–2050

Case 2 projects 2025–2050 on explicitly projected inputs (vital rates trend-extrapolated excluding COVID years, immigration fixed at 2024 levels, disaffiliation at its calibrated rate). The 2050 total is 279.7 million under both rules. The foreign-born share rises from 24.0% to 30.2% — continued immigration against fixed rates mechanically dilutes native depth — while the cap share barely moves: 0.45% $\rightarrow$ 0.42% (shallowest) and 18.35% $\rightarrow$ 16.66% (deepest). The inheritance-rule gap is thus stable over the projection horizon; demography shifts the distribution’s weight toward C_0 under either rule.

6 Conclusions and Future Developments

We have built a forward-time cluster-dynamics model of U.S. pedigree structure, grounded in measured demographic series over 1650–2025 and projected to 2050, and used it to compare two limiting definitions of inherited pedigree depth. Three conclusions stand out.

First, the inheritance rule is the dominant modeling choice. From identical births, deaths, immigration, and mating patterns, shallowest inheritance yields a 2025 America in which 0.45% of the population sits in the deepest pedigree cluster, while deepest inheritance yields 18.35% — a forty-fold gap produced by pure definition. Any empirical claim about “how deep American roots run” must therefore state its inheritance rule, or it is not a claim at all.

Second, the rule’s effect is structural, not parametric. It survives across assortativity values, across regional and religious disaggregation, and across the 2025–2050 projection; it is amplified, not created, by age structure. Totals are untouched throughout: the rule redistributes, never creates or destroys, people.

Third, the framework is validated as a forward laboratory. Hindcast error is sub-percent on totals, the α calibration is independently anchored, stochastic ensembles (being regenerated under the revised rules) test the ODE for finite-population bias, and the religious projection reproduces Pew’s measured unaffiliated share. The honest gaps — the $\sim 24\%$ total shortfall from sex-specific mortality and unauthorized accounting, the Midwest-versus-South deep-geography mismatch against self-reported ancestry — are reported as findings, since a model that cannot disagree with data cannot be trusted where data are absent.

Future developments, in order of priority: (i) age $\times$ religion joint structure, to test whether the age–deepest interaction found here concentrates further within high-endogamy groups; (ii) regional vital rates and paternal age schedules, replacing the national schedules currently imposed on every region; (iii) a settlement-history process for the colonial spin-up, replacing the all- C_0 1650 approximation; (iv) re-affiliation flows, now that disaffiliation alone is calibrated; (v) coupling the pedigree state to policy scenarios — immigration levels, fertility supports — as a compositional consequence engine for public policy.

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